



Lab-in-syringe as a practical technique for automatic microextraction: Evaluation by Blue Applicability Grade Index

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ABSTRACT

Lab-in-syringe (LIS) is a powerful automatic technique that is derived from sequential injection analysis. In LIS, a computer-controlled syringe pump is employed, and its syringe barrel serves as a mixing, reaction, and/or extraction chamber. Until now, the LIS concept has been efficiently employed for the automation of solid-phase microextraction and liquid-phase microextraction protocols as a front-end to a plethora of chromatographic and spectrometric techniques. In this work, the applicability of LIS systems was examined using the Blue Applicability Grade Index (BAGI), a recently introduced metric tool that is used to examine the applicability of an analytical method. For this purpose, the sample preparation procedure and the instrumental method are thoroughly evaluated. The attained BAGI scores ranged between 60.0 and 77.5 for all the examined methods, while the average score was 71.3. As derived from the BAGI evaluation, good practicality can be attributed to the reviewed systems and protocols.

1. Introduction

The ongoing demand for reliable, repeatable, and prompt results combined with the need for routine analysis of numerous samples has driven the development of on-line automated analytical methods employing flow injection analysis (FIA), sequential injection analysis (SIA), and related techniques. These novel methods offer various advantages such as reduced laboratory time and disposal costs, minimum consumption of chemicals, high extraction efficiency, and high enrichment factors. As a result, they have opened new avenues in the automation and miniaturization of sample preparation [1,2]. Lab-in-syringe (LIS) is a powerful concept for automated sample preparation that employs an automatic computer-controlled syringe pump. Sample preparation procedures including mixing, reaction, and/or extraction take place within the barrel of the syringe in a permanently sealed compartment [3–6]. This technique is finding increasing attention for analyte extraction and sample clean-up approaches [7], including miniaturized solid-phase extraction (SPE) and liquid-phase microextraction (LPME) [8–12].

Currently, there is a quest for developing analytical methods which, while providing highest analytical performance, at the same time offer reduced environmental impact. To achieve that, the analytical methods

must comply with the principles of Green Analytical Chemistry (GAC) [13] and Green Sample Preparation (GSP) [14]. GAC was proposed in 2000 and it concerns the role of analytical chemists in making laboratory practices more environmentally friendly by addressing the safety of solvents/reagents, toxic laboratory waste generation, the safety of workers, and the energy efficiency of an analytical method. The GSP approach focuses on sample preparation, ensuring greenness during this step and minimization of the impact on the environment and human health. Until now, multiple green metric tools have been introduced and used to evaluate the environmental impact of different analytical methods. Well-established green metric tools include the National Environmental Method Index (NEMI) [15], Analytical Eco-Scale [16], Analytical Greenness Calculator (AGREE) [17], Analytical Greenness Metric for Sample Preparation (AGREEprep) [18], Green Analytical Procedure Index (GAPI) [19], and Complementary Green Analytical Procedure Index (ComplexGAPI) [20]. Although these tools can be used to assess the green character of an analytical method, they cannot evaluate its practicality which is a basic constituent of White Analytical Chemistry (WAC) as proposed by Nowak et al. [21]. In WAC, the blue color represents productivity and practical/economic efficiency, the green color shows compliance with the principles of WAC, while the red color is assigned to analytical efficiency as expressed by validation

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criteria. To bridge this gap, the Blue Applicability Grade Index (BAGI) was recently proposed [22]. BAGI serves as a complementary tool to the previously developed green metric tools and it can be used to examine the applicability of an analytical method in terms of its practicality by method developers and users from industry, academia, and routine analysis laboratories.

BAGI is based on an open-source application (mostwiedzy.pl/bagi) and a web application available at bagi-index.anvil.app that generates an asteroid pictogram divided into different parts. These parts correspond to the type of analysis (attribute 1), the number of analytes that are simultaneously determined (attribute 2), the analytical technique (attribute 3), the number of samples that can be treated simultaneously (attribute 4), the sample preparation scheme (attribute 5), the sample throughput (attribute 6), the reagents and materials that are required (attribute 7), the fitness for purpose (attribute 8), the degree of automation (attribute 9) and the sample amount (attribute 10). Regarding the last attribute, BAGI differentiates between biological samples and food/environmental samples due to their dissimilar sample availability. A color scale of dark blue, blue, light blue, and white is used for the different parts of BAGI corresponding to diverse applicability. Fig. 1 explains the generated pictogram.

This can help scientists to identify the strong and weak points of each method. Simultaneously, the software generates a BAGI score that is shown in the middle of the pictogram. A score higher than 60 is recommended to indicate good method applicability. Thus, BAGI can be efficiently used to compare different analytical methods in terms of their practicality. The acceptance of BAGI as an efficient metric tool for sample preparation is demonstrated by the increasing number of publications that are using this tool for applicability evaluation [23–29]. Until now, the examination of the practicality of on-line LIS systems is still scarce.

In this article, the applicability of different LIS systems that have been proposed in the last decade (i.e., 2014–2024) for the automation of different sample preparation schemes was evaluated using BAGI. For the evaluation, 32 LIS systems were considered out of the 45 documents found for the Scopus search “lab-in-syringe”. The excluded papers were either developed before 2014 or did not use LIS for automatic microextraction. The main miniaturized LPME, SPE, and headspace LIS-based microextraction are discussed in terms of their practicality for sample preparation. The applicability of LIS systems based on “greener” solvents aiming to replace conventional organic solvents is also examined. In all cases, the attained BAGI scores were higher than 60.0, indicating their good practicality.

2. Evaluation of liquid-phase extraction miniaturized LIS systems

Until now, LIS has been efficiently used for the automation of various LPME systems that have been proposed as a front-end to chromatographic techniques such as high performance liquid chromatography (HPLC) [30] and gas chromatography (GC) [31], as well as

spectrometric techniques such as inductively coupled plasma optical emission spectrometry (ICP-OES) [32], flame atomic absorption spectrometry (FAAS) [33], electrothermal atomic absorption spectrometry (ETAAS) [34], and cold vapor atomic absorption spectrometry (CVAAS) [35].

2.1. Evaluation of dispersive liquid-liquid microextraction LIS systems

In the conventional format of dispersive liquid-liquid microextraction (DLLME), an extracting solvent, a dispersing solvent and an aqueous sample are mixed, to produce a cloudy solution that contains numerous droplets of the extracting medium. Due to the high surface area of the solvent droplets, the target analytes are transferred quickly into a fine emulsion of the extractant, and equilibrium is rapidly attained. Centrifugation is then employed to improve the separation of the analyte-rich phase to be used for analysis. Chloroform, dichloroethane, and tetrachloromethane are common extractants (mostly chosen for their immiscibility with water and greater density), while acetonitrile and methanol are typically used as dispersants [36]. Table 1 presents the results of the evaluation of the practicality of different LIS systems developed for the automation of LPME approaches.

Caffeine has been extracted from coffee beverages by automatic DLLME using a LIS system coupled with spectrophotometry [37]. In this case, an aliquot of 3 mL of sample was required, dichloromethane was used as extractant, and methanol as disperser solvent. Since the extractant is heavier than water, the syringe module was used upside-down. This procedure showed multiple advantages since it was based on common commercially available reagents, and it resulted in sufficient sensitivity and high sample throughput (e.g., 16 h^{-1}). The attained BAGI score was 75.0, showing great practicality. This score can be further improved through the incorporation of more analytes in the analytical scheme and the utilization of instrumentation that can result in both quantitative and confirmatory results. In most approaches spectrophotometry was considered common equipment that is readily available in QC laboratories. In the case of portability, the overall BAGI score is expected to increase by 2.5 points.

An upside-down format was also proposed for the stirring-assisted DLLME of methylene blue active substance index in environmental and drinking water samples using chloroform as extractant [38]. The principle of this method is based on the formation of ion pairs among methylene blue and ionic surfactants, which are determined after extraction and spectrophotometric determination. Since sodium dodecyl benzene sulphonate was used as standard solution of anionic surfactants, single-parameter analysis was considered for the first criterion of BAGI. The proposed approach required 4 mL of sample and the entire procedure required approximately 345 s resulting in a sample throughput of 10 h^{-1} . A score of 75.0 was obtained, showing good practicality.

A similar approach was proposed for the determination of cationic surfactants as an ion-pair with disulfine blue dye [39]. Chloroform was employed as an extraction solvent and an upside-down syringe

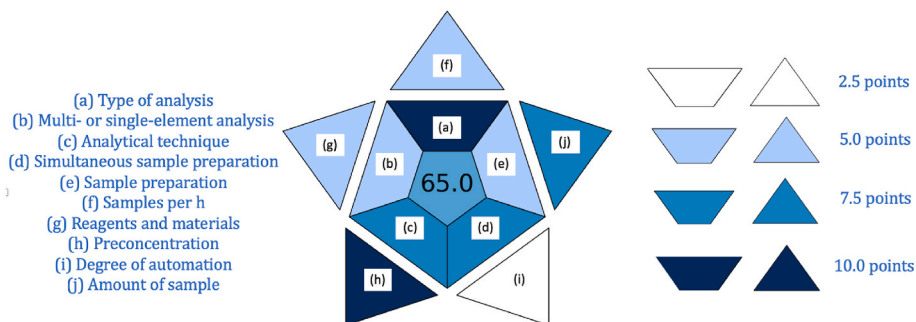


Fig. 1. BAGI pictogram explained.

Table 1
Practicality evaluation of different LIS systems developed for the automation of LPME.

Analyte	Instrumentation ^a	Sample type/volume	Reagents and materials	Sample preparation	Samples per h	BAGI pictogram	BAGI score	Ref.
Caffeine	Spectrophotometry	Coffee beverages (3 mL)	Common commercially available reagents	Dispersive liquid-liquid microextraction	16		75.0	[37]
Anionic surfactants	Spectrophotometry	Drinking and environmental water (4 mL)	Common commercially available reagents	Dispersive liquid-liquid microextraction	10		75.0	[38]
Cationic surfactants	Spectrophotometry	Drinking and environmental water (4 mL)	Common commercially available reagents	Dispersive liquid-liquid microextraction	10		72.5	[39]
Pb	Spectrophotometry	Industrial residual water (1.25 mL)	Common commercially available reagents	Dispersive liquid-liquid microextraction	13		77.5	[40]
Cd, Cu, Pb, Ag	ICP-OES	Food and environmental samples (<10 mL)	Common commercially available reagents	Dispersive liquid-liquid microextraction	9		75.0	[41]
Nitrophenols (n = 3)	Spectrophotometry	Surface water (9 mL)	Common commercially available reagents	Dispersive liquid-liquid microextraction	5		75.0	[42]
		Surface water (24 mL)			3		70.0	
Silver	ETAAS	Environmental water (4 mL)	Common commercially available reagents	Liquid phase microextraction	15		77.5	[34]
Pb	Spectrometry	Drinking water (3.3 mL)	Common commercially available reagents	Direct immersion single drop microextraction	10		72.5	[43]
		Drinking water (3 mL)		In drop-stirring microextraction	12		75.0	
Beta-blockers (n = 4)	UHPLC-MS/MS	Serum (100 µL)	Common commercially available reagents	Protein precipitation and salting-out LLE	7		70.0	[30]
Sulfonamides (n = 4)	HPLC-UV	Urine (<100 µL diluted 1:5)	Commercially available reagents not common in QC labs	Salting out liquid-liquid extraction step and an online SPE	4.4		67.5	[44]
Sulfonamides (n = 4)	HPLC-UV	Milk (500 µL)	Common commercially available reagents	Protein precipitation and salting-out LLE	Approximately 6		70.0	[45]

(continued on next page)

Table 1 (continued)

Analyte	Instrumentation ^a	Sample type/volume	Reagents and materials	Sample preparation	Samples per h	BAGI pictogram	BAGI score	Ref.
Pesticides (n = 4)	HPLC-MS/MS	Fruit and berry juice (300 μ L)	Common commercially available reagents	Sugaring-out liquid-liquid extraction	Approximately 4		72.5	[46]
Antimony	Spectrophotometry	Environmental water, antileishmanial drugs	Commercially available reagents not common in QC labs	Cloud point extraction	16		72.5	[47]
Copper	ETAAS	Vegetable oils (4 mL)	Needs to be synthesized (simple way)	Liquid phase microextraction	5		70.0	[49]
Sudan dyes (n = 3)	HPLC-DAD	Chilli sauce (1 g)	Needs to be synthesized (simple way)	Liquid phase microextraction	Approx. 5		72.5	[50]
Fluoroquinolones (n = 5)	HPLC-FLD	Surface water and wastewater treatment plant effluent (3 mL)	Needs to be synthesized (simple way)	Single drop microextraction	3		70.0	[52]
Sulfonamides (n = 2)	HPLC-UV	Human urine (100–500 μ L)	Needs to be synthesized (simple way)	SHS microextraction	<2		62.5	[56]

^a ICP-OES: Inductively coupled plasma-optical emission spectroscopy, ETAAS: Electrothermal atomic absorption spectroscopy, UHPLC-MS/MS: Ultra high performance liquid chromatography-tandem mass spectrometry, HPLC-UV: High performance liquid chromatography-ultraviolet detection, UHPLC-MS/MS: High performance liquid chromatography-tandem mass spectrometry, HPLC-DAD: High performance liquid chromatography-diode array detection, HPLC-FLD: High performance liquid chromatography-fluorescence detection.

orientation was employed. The disulfine blue active substances index was expressed as concentration of cetyltrimethylammonium bromide (single analyte for the second attribute of BAGI). The method was proposed for water sample analysis, and an aliquot of 4.0 mL of sample was required. After the DLLME procedure, the new system configuration enabled extract washing with water and barium acetate solution to minimize interferences. A sample throughput of 6.6 h⁻¹ was achieved in this case. This LIS system attained a BAGI score of 72.5, proving its high practicality.

A LIS flow system exploiting DLLME combined with spectrophotometric determination has been proposed for the quantification of lead in industrial residual waters [40]. Xylene was used as extractant as a “greener” alternative to replace the traditional solvents used in DLLME. A portable system was generated which is a noteworthy feature of the proposed system. The sample throughput was estimated as 13 h⁻¹. Dithizone was employed as complexing agent and chromogenic reagent for the spectrophotometric determination. Reduced sample consumption (i.e., 1250 μ L) was required, which was a benefit of the novel system that attained a BAGI score of 77.5.

For multi-elemental analysis, a fully automated LIS system has been proposed for the magnetic stirring-assisted DLLME hyphenated to ICP-OES [41]. A high-temperature torch integrated sample introduction system was employed, and direct injection of the organic phase was achieved. Xylene was used as the extraction solvent and the target analytes were extracted as diethyldithiophosphate complexes. Since this introduction system is not common in quality control laboratories, sophisticated instrumentation was selected for the third attribute of BAGI. The proposed method was used for the analysis of environmental (i.e., seawater) and food samples (i.e., salt, salt without sodium, grape juice, and apple juice). A sample throughput of 9 h⁻¹ was achieved. This

system attained a BAGI score of 75.0, showing high practicality.

Continuous DLLME in flow-batch has been reported by Fikarova et al. [42]. Through a flow channel in the syringe piston, continuous flow through the syringe void was enabled resulting in an automatic system that can be used for adaptable sample amounts. 1-Octanol was used as extraction solvent and dispersion was achieved by the action of a magnetic stirring bar. The extraction procedure was combined with dispersive back-extraction in an alkaline aqueous acceptor phase. The obtained system was used for the determination of three mono-nitrophenols (i.e., o-, m-, p-nitrophenol) as priority pollutants in surface water samples. To attain tunable sensitivity, validation of the developed method was conducted using 9 mL and 24 mL of sample, resulting in a sample throughput of 5 and 3 h⁻¹, respectively. The attained BAGI scores ranged between 70.0 (for a sample consumption of 24 mL) and 75.0 (for a sample consumption of 9 mL). Thus, good method practicality is achieved for the different sample consumptions.

2.2. Evaluation of other LPME LIS systems based

Silver has been determined in environmental water samples using an automatic stirring-assisted liquid-liquid microextraction LIS system [34]. In this approach, the metal ion reacted with the chelating reagent, and the metal complex was extracted by diisobutyl ketone under stirring, followed by its subsequent transportation and determination by ETAAS. An aliquot of 4 mL of sample was required, the procedure required only common commercially available reagents, and the sample throughput was 15 h⁻¹. The utilization of ETAAS resulted in both quantitative and confirmatory results. A BAGI score of 77.5 was attained, proving the practicality of the proposed system for trace metal analysis.

Šrámková et al. [43] developed a direct-immersion single-drop microextraction LIS system and an in-drop stirring microextraction LIS system for the determination of lead in drinking water. Both methods build on the spectrophotometric dithizone assay determination of the target analyte. The former procedure involved the extraction of the complex in a floating drop of a toluene-hexanol mixture pumped using normal syringe orientation, while an upside-down orientation was used for the latter approach and chloroform was used as extraction solvent. Enhanced extraction efficiency was obtained for the in-drop stirring microextraction by placing a magnetic stir bar inside the syringe for homogenous mixing. The required sample amount was less than 3.3 mL in both cases. The resulting sample throughput was 10 h⁻¹ for the direct-immersion single-drop microextraction and 12 h⁻¹ for in-drop stirring microextraction, resulting in BAGI scores of 72.5 and 75.0 respectively. Thus, apart from the lower detection limit, the in-drop stirring approach resulted in higher practicality.

The determination of four beta-blockers in serum following automated protein precipitation and salting out liquid-liquid extraction has been reported by Yıldırım et al. [30]. The sample preparation was achieved through the aspiration of 100 µL of sample and 200 µL of acetonitrile to achieve protein precipitation, followed by the aspiration of a salt solution to induce phase separation for homogeneous liquid-phase microextraction. Since these procedures took place simultaneously, a miniaturized extraction scheme was selected for the fifth criterion of BAGI. The extracted analytes were contained in the resulting floating organic layer, an aliquot of which was further diluted and analyzed. The LIS system was coupled online with UHPLC-MS/MS resulting in both quantitative and confirmatory results. By performing the sample preparation and the chromatographic run in parallel, a sample throughput of 7 h⁻¹ was achieved. Among the benefits of this system is the reduced sample amount (i.e., 100 µL) that was required and the utilization of common commercially available reagents, resulting in a BAGI score of 70.0. Thus, despite the complexity of the sample matrix and the increased complexity of serum samples, a highly practical system was developed.

Two automated LIS systems have been developed for the determination of sulfonamides in urine [44] and milk samples [45]. The analysis of urine samples was performed after the double-stage sample preparation that included a salting out liquid-liquid extraction step and an online solid-phase extraction step [44]. The homogeneous liquid-liquid extraction was performed at pH 3 using water-miscible acetonitrile, while a saturated solution of kosmotropic salts MgSO₄ and NaCl induced the phase separation. Accordingly, the analytes were trapped in an anion-exchange resin cartridge integrated into the HPLC injection loop (pH 10) to improve the overall method clean-up and preconcentration. The quantification of the four analytes was performed by HPLC-UV, which is a common instrumentation that results in quantitative results. The protocol required commercially available reagents not common in laboratories (i.e., SPE cartridges) and reduced sample consumption. Good sensitivity was achieved eliminating the need for further preconcentration. A BAGI score of 67.5 was obtained, demonstrating good practicality. A centrifugation-less deproteinization and homogenous liquid-liquid extraction of four sulfonamides has been proposed for milk analysis using LIS [45]. In this case, simultaneous salting-out liquid-liquid extraction of the analytes and sample deproteinization was performed. After the miniaturized extraction scheme, the samples were *in situ* diluted and online transferred for HPLC-UV analysis. Analyte separation and the subsequent extraction were performed in parallel resulting in a sample throughput of approximately 6 h⁻¹. The procedure was similar to the one reported for urine sample analysis, and it required only 500 µL of milk. The miniaturized extraction sample preparation scheme was selected for the fifth criterion of BAGI since it was performed simultaneously with protein precipitation. An overall score of 75.0 was achieved, demonstrating the applicability of this system. This can be attributed to the utilization of common commercially available reagents, the high obtained sensitivity, and the low sample

consumption.

The miniaturization and automation of sugaring-out liquid-liquid extraction using the LIS system combined with HPLC-MS/MS had been reported for the determination of pesticides in fruit and berry juices [46]. Fig. 2 shows the proposed manifold. Four analytes belonging to different pesticide classes were extracted in water-miscible organic solvent acetonitrile followed by phase separation using glucose as sugaring-out reagent. Among the benefits of the proposed method are the reduced sample consumption (i.e., 300 µL) and the diverse range of analytes included in the study. Although the LIS system was operated in less than 2 min, the sample analysis time was 13 min (including equilibration), resulting in a sample throughput of approximately 4 h⁻¹. A BAGI score of 72.5 was attained, demonstrating good practicality. This can be further improved by the reduction of sample analysis time (e.g., using UHPLC instead of conventional HPLC).

Finally, an analytical procedure exploiting the cloud point extraction on a LIS flow system for the spectrophotometric determination of antimony in environmental water and antileishmanial drugs has been proposed by Frizzarin et al. [47]. The sample preparation procedure was based on the formation of anion-pair among antimony-iodide complex, followed by its extraction using Triton X-114. Thus, this system required the utilization of commercially available reagents not common in QC labs. A sample throughput of 16 h⁻¹ was achieved and the consumption of 2 mL of sample solution was required. High overall method sensitivity was obtained, and thus no further preconcentration was required. A BAGI score of 72.5 was obtained which can be further improved by incorporating more analytes in the analytical scheme and through the selection of instrumentation that results both in quantitative and confirmatory information.

2.3. Evaluation of LPME LIS systems based on green solvents

Recently, the replacement of conventional organic solvents with “greener” alternatives and their utilization in LIS miniaturized extraction systems has attracted the interest of researchers. Deep eutectic solvents (DES) are compounds formed by a donor and an acceptor capable of hydrogen bond formation, which are considered to be an environmentally friendly and cheap alternative to ionic liquids. DES are in a liquid state at ambient temperature, and they show multiple desirable properties such as non-volatility, and high thermal stability. In analytical chemistry, they have proved to be effective extractants of a wide range of organic and inorganic analytes [48].

An automated reversed-phase liquid-liquid microextraction based on deep eutectic solvent (DES) has been proposed by Shishov et al. [49] for the determination of copper in vegetable oil samples. For this purpose,

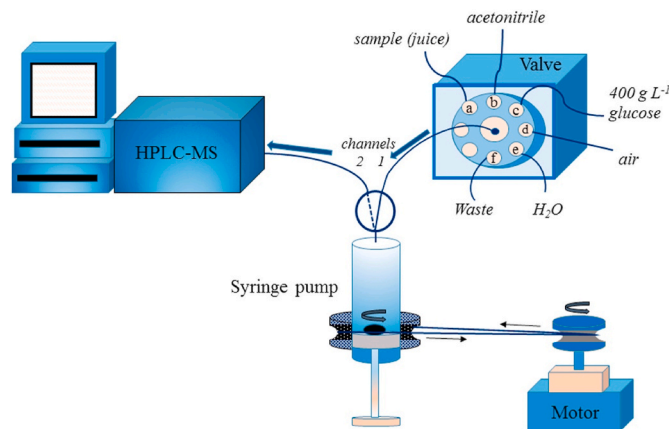


Fig. 2. On-line sugaring out LIS system combined with HPLC-MS for pesticide determination in juice samples. Reproduced with permission from Ref. [46] Elsevier. Copyright Elsevier, 2024.

an aliquot of 4 mL of sample was mixed with the extractant which was prepared by mixing choline chloride, lactic acid, and water. Lactic acid also served as the chelating agent for copper determination. Water was employed to decrease the viscosity of the extractant. Extraction was performed under mixing into the syringe pump. After the spontaneous phase separation, the analyte was determined by ETAAS. The proposed system exhibited a sample throughput of 5 measurements per hour. A BAGI score of 70.0 was attained, which can be further enriched by the incorporation of more analytes in the analytical scheme.

Natural deep eutectic solvents (NADESs) have been used in automated LIS systems to replace conventional solvents and enhance the green character of microextraction [50]. NADESs are alternative to DESs that consist of naturally occurring compounds such as amino acids, terpenes, organic acids, and sugars among others. They are characterized by sustainability, simplicity in preparation, reduced cost, negligible volatility, adjustable viscosity, and good extraction performance towards different analytes [51]. A NADES made of menthol and hexanoic acid (1:2) was used for the extraction of three Sudan dyes from chili sauce samples prior to their determination by HPLC-DAD. The microextraction procedure was performed within 2 min under mixing. Subsequently, the mixing was stopped for 3 min and the extract was transferred to an HPLC vial. Using an autosampler this approach can be fully automated. Since the HPLC run time was approximately 10 min, a sample throughput of ca. 5 h⁻¹ can be achieved. The BAGI score of this approach was found to be 72.5, showing good practicality due to the reduced sample amount (i.e., 1 g of sauce), the high obtained sensitivity, and the automation of the procedure. A disadvantage of this protocol is that the NADES is not commercially available, however, it can be prepared in the laboratory in a simple way.

The automation of NADES solvent-based direct immersion single drop microextraction using LIS has been also reported [52]. In this case, a solvent synthesized from hexanoic acid and thymol was used as extractant and the obtained system was proposed for the determination of five fluoroquinolones combined with high performance liquid chromatography-fluorescence detection (HPLC-FLD). For the extraction of the target analytes, an aliquot of 3 mL of sample and a buffer solution were aspirated in the syringe barrel, and a drop of NADES (60 µL) was exposed. A small air volume was aspirated to create a “bubble-in-drop” and inflate the solvent, enhancing its interfacial area. Extraction was achieved within 16 min, and the total procedure lasted 20 min enabling the parallel sample extraction and analysis. Thus, a sample throughput of 3 h⁻¹ was achieved. The proposed method was used for the analysis of surface water and wastewater treatment plant effluent. A BAGI score of 70.0 was attained, demonstrating the good applicability of the proposed system.

Apart from DESs and NADESs, supramolecular solvents (SUPRASS) have been used in LPME protocols to provide “greener” alternatives. SUPRASSs are water-immiscible liquids composed of amphiphile aggregates. They are generated through sequential self-assembly, induced by the application of a trigger such as changes in pH, temperature, or salt addition [53]. The automatic surfactant-assisted DLLME of thirteen polycyclic aromatic hydrocarbons from tea infusions in SUPRAS based on a LIS system has been proposed. The procedure was performed by mixing the aqueous sample phase and a hydrophilic emulsion containing hexanoic acid and sodium hexanoate that acted as an emulsifier and that was required for the formation of the SUPRAS phase. The hydrophilic emulsion was prepared in a simple way by mixing hexanoic acid and sodium hydroxide. The quantification of the target analytes was performed by HPLC-FLD. Although the microextraction protocol was rapidly performed resulting in a sample preparation throughput of 15 h⁻¹, the separation of the analytes was performed in 18 min, resulting in an overall sampling frequency of approximately 3 h⁻¹. For the sample preparation, an aliquot of 1 g of tea leaves was brewed in 10 mL of hot water, and 3.8 mL of the resulting solution was analyzed. A BAGI score of 72.5 was attained, showing good method practicality while it efficiently replaced the use of conventional organic solvents.

Switchable-hydrophilicity solvents (SHS) are another example of environmentally friendly “smart” solvents that have been used for the development of automatic systems. They include amines, fatty acids, solvent mixtures, and various surfactants. During extraction processes, SHSs can be interchanged between hydrophobic and hydrophilic forms by adding a “trigger” (i.e., carbon dioxide, pH change). These solvents can be used for the extraction of organic and inorganic compounds in various complex matrices and mixtures, combined with chromatographic, spectroscopic, and electroanalytical techniques [54,55]. Pochivalov et al. [56] proposed a fully automated LIS switchable hydrophilicity solvent-based microextraction approach using di-(2-ethylhexyl)phosphoric acid as SHS. Although the SHS was commercially available, a purification procedure was required to be performed in the laboratory using common instrumentation. The microextraction protocol was combined with HPLC-UV for the determination of two sulfonamides in human urine samples. The required sample amount fell within the range 100–500 µL for biological samples. A sample preparation frequency of 12 h⁻¹ was achieved; however, the total analysis time was 35 min including equilibration, resulting in a total sample throughput of less than 2 h⁻¹. Although the LIS system was not on-line coupled to the chromatograph and a dilution of the extract was performed in an Eppendorf tube, the use of HPLC autosampler can result in a fully automatic system for sample preparation and analysis. A BAGI score of 62.5 was attained, demonstrating that the obtained system is practical. This score can be further increased through the incorporation of more analytes in the method, the reduction of sample amount, the reduction of the sample analysis time, and the selection of commercially available reagents that can be used as such.

2.4. Evaluation of headspace miniaturized LIS systems

BAGI was also employed for the evaluation of different headspace miniaturized LIS systems (Table 2). A stirring-assisted room-temperature headspace analysis of benzene, toluene, ethylbenzene, and xylenes in surface waters has been proposed by Horstkotte et al. [31]. This system was combined on-line with GC-FID resulting in quantitative results for all analytes. The syringe pump of the LIS system was used upside-down to have sufficient headspace between the sample and the piston and their partitioning to the headspace was carried out under homogeneous mixing using a magnetic stirring bar. The analysis time was less than 19 min, resulting in the analysis of less than 4 h⁻¹. Among the benefits of this approach are the reduced sample amount (i.e., 1.8 mL), the utilization of common commercially available chemicals, the good sensitivity that eliminated the need for further preconcentration, the simple low-cost extraction scheme, and the wide range of analytes included in the study. Thus, a BAGI score of 75.0 was attained, corresponding to great practicality.

The automation of head-space single-drop micro-extraction (HS-SDME) has been reported by Šrámková et al. for the determination of ethanol in wine samples [57] and Mitani et al. for the determination of mercury in environmental water samples [58]. In the former approach, the drop was positioned in the syringe inlet in the headspace above the sample with posterior spectrophotometric quantification, and the application of negative pressure during the enrichment step was possible, favoring the evaporation of the volatile analyte. The extraction was performed under mixing using a magnetic stir bar. Among the benefits of this approach were the reduced sample volume (i.e., 0.1 mL of wine diluted to a final volume of 1.0 mL), the high sample throughput (i.e., 12 samples per h for one repetition), the simple required instrumentation (i.e., spectrophotometer), the high obtained method sensitivity, and the common commercially available reagents that were required. This resulted in high method practicality, as reflected by the BAGI score of 75.0 that was attained [57]. For mercury determination by HS-SDME, an aliquot of 3.5 mL of sample was aspirated within the syringe barrel together with a SnCl₂ solution (reductant) and air. Pre-concentration of the mercury reduced *in situ* and Hg⁰ took place into a

Table 2

Practicality evaluation of different LIS systems developed for the automation of headspace sample preparation techniques.

Analyte	Instrumentation ^a	Sample type/ volume	Reagents and materials	Sample preparation	Samples per h	BAGI pictogram	BAGI score	Ref.
Benzene, toluene, ethylbenzene, xylenes	GC-FID	Surface water (1.8 mL)	Common commercially available reagents	Headspace extraction	2–4		75.0	[31]
Ethanol	Spectrophotometry	Wine (0.1 mL diluted to 1 mL)	Common commercially available reagents	Single drop headspace extraction	12		75.0	[57]
Mercury	ETAAS	Wine (0.1 mL diluted to 1 mL)	Common commercially available reagents	Single drop headspace extraction	8		75.0	[58]
Ammonia	Fiber optics spectrophotometry	Surface water (<10 mL)	Common commercially available reagents	Single drop headspace extraction	17		77.5	[59]
Ammonia	MiniSIA	Environmental water (3 mL)	Commercially available reagents not common in QC labs	Pressure-assisted dual- headspace gas-liquid microextraction	8		70.0	[60]

^a GC-FID: Gas chromatography-flame ionization detector, ETAAS: Electrothermal atomic absorption spectrometry, MiniSIA: Mini Sequential Injection Analysis.

25 μ L microdrop of palladium solution under stirring at reduced headspace pressure. Afterward, ETAAS was employed for the quantification of the target analyte. The whole procedure was fully automated, and it required common commercially available reagents. The enhanced sample throughput of 8 samples per h, the possibility to obtain both quantitative and confirmatory results, and the reduced sample amount were among the benefits of this system, which attained a BAGI score of 75.0, proving its practicality for mercury determination [58]. An upside-down orientation for headspace in-syringe magnetic stirring was proposed for the LIS HS-SDME of ammonia from surface water samples [59]. This approach enabled the homogenous solution mixing, the promotion of the head-space enrichment of the target analyte, and the efficient syringe cleaning. This approach required a modification of the syringe piston which was drilled and a PEEK tube was then inserted through the generated hole enabling the microdrop used for SDME to hang vertically on the piston orifice. The extraction of ammonia took place on a drop of bromothymol blue indicator, while on-drop sensing the color change via fiber optics. The selection of simple-to-operate instrumentation served as a significant advantage of this protocol. Moreover, the obtained sample throughput was high (i.e., 17 h⁻¹), enhancing the practicality of this method. A BAGI score of 77.5 was obtained, which was higher than the other two HS-SDME systems, demonstrating the superiority of this approach.

The determination of ammonia from environmental water samples has been also reported using a fully automated pressure-assisted dual-headspace LIS system [60]. In this case, the *in-situ* vapor generation of ammonia was carried out under low pressure in the headspace of a micro-syringe. Subsequently, the vapor was on-line transferred into the headspace of another micro syringe to facilitate the dissolution of gaseous ammonia under positive pressure. After the dissolution of ammonia, a fluorescent product was formed, and its quantitative determination was performed spectrophotometrically using a portable miniSIA system. An aliquot of 3 mL of water was required and a sampling frequency of 8 h⁻¹ was achieved. A BAGI score of 70.0 was attained due to the reduced sample consumption, the high method sensitivity, and the simple and portable instrumentation required.

An average score of 72.7 was attained for all the examined LPME and headspace microextraction systems, showing the practicality of using LIS for the automation of these approaches.

3. Evaluation of solid-phase extraction miniaturized LIS systems

Until now, the number of LPME LIS systems has prevailed over SPE miniaturized LIS systems. However, the recent advances in nanotechnology together with the demands for the development of novel sorbents in sample preparation have resulted in the development of automated LIS systems based on various novel sorbents. Table 3 summarizes the results of the evaluation of the practicality of different LIS systems developed for the automation of solid-phase extraction approaches. Initially, the BAGI tool was used to evaluate the practicality of three foam-based microextraction LIS systems that were proposed for the automatic determination of inorganic [33,35] and organic analytes [61]. Foam or sponge sorbents have been used for the development of automated sample preparation techniques due to their favorable characteristics including high compressibility, high surface area due to their open-hole structure, negligible weight, good extraction efficiency, and reusability. Polyurethane foams (PUFs) were among the first sorbents that have been employed for microextraction foams, while the development of new foam materials has attracted the interest of many researchers to take advantage of their beneficial properties [62,63].

Toxic metals have been determined in environmental water samples by LIS-based continuous flow-foam solid phase microextraction [33]. Fig. 3 shows the developed manifold. In this case, polyurethane foam was used as sorbent material for the adsorption of cadmium and lead after on-line complexation with ammonium pyrrolidine dithiocarbamate, while the elution was performed using methyl isobutyl ketone (400 μ L). FAAS was employed as an analytical technique resulting in both quantitative and confirmatory results, while an aliquot of 12 mL of water sample was required. Among the benefits of the proposed method are the high sample throughput (18 h⁻¹) that was achieved, the utilization of common commercially available reagents, and the high obtained sensitivity that eliminated the need for additional sample

Table 3

Practicality evaluation of different LIS systems developed for the automation of solid-phase extraction approaches.

Analyte	Instrumentation ^a	Sample type/volume	Reagents and materials	Sample preparation	Samples per h	BAGI pictogram	BAGI score	Ref.
Cd, Pb	FAAS	Water (12 mL)	Common commercially available reagents	Foam microextraction	18		77.5	[33]
Hg	CVAAS	Biological samples (digested)	Commercially available reagents not common in QC labs	Foam microextraction	14		67.5	[35]
BPA	HPLC-UV	Drinking water (50 mL)	Needs to be synthesized (simple way)	Foam microextraction	8		67.5	[61]
		Food simulant (10 mL)			<2		62.5	
Cd, Pb, and Cu	ICP-OES	Biological samples (100–500 mg)	Needs to be synthesized (simple way)	Fabric disk extraction	8		60.0	[64]
		Food and environmental samples (<10 mL)					62.5	
Malachite green	Spectrophotometry	Environmental water, drinking water, fish samples (10 mL)	Needs to be synthesized with advanced know-how	Dispersive microsolid-phase extraction using MOFs	18		67.5	[66]
Water contaminants (n = 5)	HPLC-UV	River and lake water (7 mL)	Needs to be synthesized (simple way)	Dispersive microsolid-phase extraction	Approx. 4.5		72.5	[12]

^a FAAS: Flame atomic absorption spectrometry, CVAAS: Cold vapor atomic absorption spectrometry, HPLC-UV: High performance liquid chromatography-ultraviolet detection, ICP-OES: Inductively coupled plasma-optical emission spectroscopy.

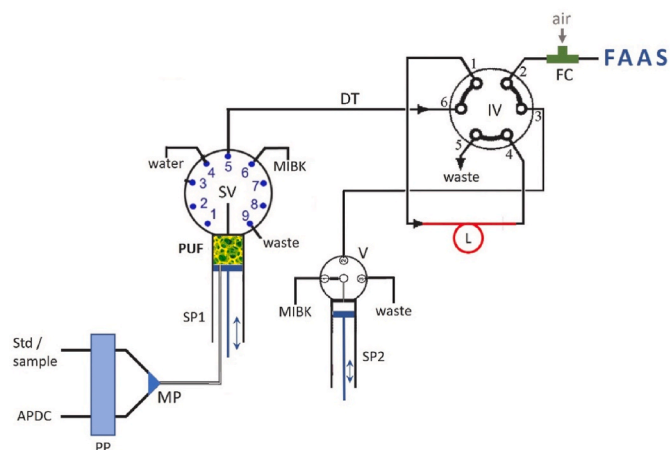


Fig. 3. A continuous flow polyurethane foam SPE-based LIS system combined with FAAS for toxic metal determination. Reproduced with permission from Ref. [33] Elsevier. Copyright Elsevier, 2017.

treatment. A BAGI score of 77.5 was achieved showing good method practicality.

A polyurethane foam-based microextraction LIS system has been also proposed for the determination of mercury in environmental water samples [35]. In this approach, a cylindrical plug of PUF was immobilized in the bottom of the syringe, directly at the piston which was drilled enabling the insertion of PTFE tubing through it. Using this configuration, argon could be directly purged in the syringe, while the continuous loading of the sample solution was enabled using a peristaltic pump. After the on-line retention of the analyte, *in situ* reduction was achieved using a HCl/SnCl₂ solution, and the vapor was transferred to CVAAS for mercury quantification. In this case, the attained BAGI score was 67.5, revealing good method applicability, which could be further improved through the reduction of the required sample volume.

Sol-gel poly(tetrahydrofuran) coated foam microextraction combined with HPLC-UV has been employed for the determination of bisphenol A (BPA) in drinking water samples and food simulants in migration studies from polycarbonate plastic materials [61]. For this purpose, the foam was cut into cylindrical plugs, and it was immobilized at the top of a glass barrel of a micro-syringe pump. During sample aspiration and dispensation, the sample solution could easily penetrate the pores of the sorbent, which could be easily compressed due to its

inherent properties. In this case, the quantitative determination of one compound could be performed in a fully automatic approach. After the LIS procedure, no additional treatment was required. A disadvantage of this procedure is that the foams were not commercially available, however, their preparation was performed in a simple way using common instrumentation. To obtain the required sensitivity for migration studies, 10 mL of sample was required, resulting in a sample throughput of 8 h^{-1} , while a lower sample throughput was obtained for drinking water samples that required a sample consumption of 50 mL. Thus, BAGI scores of 62.5 and 67.5 were attained for drinking water analysis and food simulant analysis, respectively. Thus, good practicality was demonstrated in both cases, which could be further improved with the incorporation of more analytes in the analytical method for their simultaneous determination with BPA.

Trace metals have been also determined in environmental, biological, and food samples using a LIS system based on the on-line dual lab-in-syringe flow-batch fabric disk sorptive extraction (FDSE) platform combined with back-extraction [64]. FDSE is the automated format of fabric phase sorptive extraction which was originally introduced by Kabir and Furton in 2014 [65]. In this case, permeable and flexible fabric substrates coated with a sol-gel organic-inorganic sorbent are used for the extraction of different target analytes. Among the benefits of this technique are its high extraction efficiency, the tunable selectivity of the extraction membrane, and the rapid extraction kinetics [32]. For the assembly of the dual LIS system, sol-gel coated poly(caprolactone)-poly(dimethylsiloxane)-poly(caprolactone)-coated polyester fabric disks were packed at the top of the glass barrel of the first syringe pump, while a back-extraction chamber was incorporated on a second syringe pump. For the determination of the target analytes, the metals were extracted after their complexation with ammonium diethyldithiophosphate, eluted by diisopropyl ketone, and subjected to oxidative back-extraction prior to their determination by ICP-OES. A high sampling frequency of 8 h^{-1} was achieved, resulting in a blue color of the respective criterion. The preconcentration of the analyte was achieved after multi-step sample preparation. An overall BAGI score of 62.5 was attained for food and environmental samples since less than 10 mL of sample were required, which could be further improved by reducing this amount. For biological samples, an aliquot between 100 and 500 mg was used for digestion, resulting in a BAGI score of 60.0. Thus, good method practicality was achieved taking into consideration all the different sample types.

A dispersive micro solid phase extraction (μ -SPE) method employing magnetic metal-organic frameworks (MOFs) based on a LIS concept was proposed combined with spectrophotometry for the determination of malachite green [66]. MOFs are porous hybrid materials composed of metal ions and organic linkers. Due to their beneficial characteristics such as high surface area, they have proved to be beneficial sorbent materials in extraction and microextraction procedures [67]. The magnetic sorbent was retained on the surface of a miniature magnetic bar that was placed inside of the automatic syringe. In this way, the activation of magnetic stirring enabled sorbent dispersion, while its deactivation enabled the magnetic retrieval of the material. The proof-of-concept of this system was demonstrated for drinking water, environmental water, and fish sample analysis. The analysis throughput was 18 h^{-1} , while for sample analysis less than or equal to 10 mL could be used. A BAGI score of 67.5 was attained, showing the practicality of the novel system. A disadvantage of this system is that the MOF sorbent needs to be synthesized in the laboratory using advanced equipment and know-how.

A renewable sorbent μ -SPE protocol automated by LIS was proposed by Gemuh et al. [12] in 2022. In this case, a simple procedure was adopted for the functionalization of commercial Supel™-Select HLB (Hydrophilic modified styrene polymer) sorbent beads allowing the embedding of Fe_3O_4 nanoparticles. This material was used for the automatic LIS extraction of surface water contaminants that belonged to five different analyte classes. The LIS system was coupled online to

HPLC-UV and it was demonstrated for river and lake water analysis. For the automatic extraction, an aliquot of 7 mL was required to provide the required method sensitivity. No additional preconcentration was then necessary, which ensured the practicality of this approach. In this approach, a sample throughput of approximately 4.5 h^{-1} was achieved due to the HPLC run time of 13 min. The parallel operation of the HPLC and LIS systems was performed and thus $5\text{--}10 \text{ h}^{-1}$ was chosen for the sixth attribute of BAGI after rounding. An overall score of 72.5 was achieved, showing good method practicality.

All things considered, the eight SPE-based LIS systems that were reported between 2014 and 2024 showed an average BAGI score of 67.2. Thus, the incorporation of solid sorbents and supports within the glass barrel of the LIS system can result in practical systems that can serve as efficient tools for QC laboratories that work on the analysis of a high number of samples in a short time. A limitation of most of the examined systems is the fact that they are based on non-commercially available sorbents. Thus, in terms of practicality, it is more desirable to develop systems based on common sorbents assuming that they will result in the development of a system that meets the desired performance characteristics.

4. Conclusions

In this work, the recently developed BAGI metric tool was employed to evaluate the applicability of solid-phase extraction miniaturized LIS systems and liquid-phase microextraction LIS systems that were developed in the last decade. BAGI takes into consideration the “blue” principles of white analytical chemistry, and it can be used to provide both qualitative and quantitative results regarding method practicality. BAGI was used for the evaluation of 32 LIS systems. Among them, eight systems were based on solid-phase extraction, eighteen were based on LPME, five were based on headspace extraction miniaturized, and one system was based on the combination of SPE and LPME. The attained scores ranged between 60.0 and 77.5, while the average score was 71.3. The narrow range of the scores can be attributed to the inherent characteristics of the LIS systems that were developed (i.e., similar sample consumption, coupling with common analytical techniques, elimination for the requirement of sample preconcentration, high sample throughput etc.). The average score for LPME and headspace microextraction systems was 72.7, while the SPE-based LIS systems showed an average BAGI score of 67.2. The higher score of LPME methods can be attributed to the utilization of common commercial reagents in the majority of these approaches, while novel extraction phases were used in most of the SPE-based LIS systems. However, when the selection of an analytical method is done the researcher should also consider its performance characteristics and its green character, since most LPME require the use of more toxic solvents.

Since all of the examined systems showed a BAGI score higher than 60.0, good method applicability was demonstrated for LIS approaches. On the whole, LIS serves as a practical automatic technique that can be combined with different chromatographic and spectrometric methods for the determination of both organic and inorganic analytes in complex samples. Based on the evaluation of the reported systems using the BAGI metric tool, it is generally recommended to include a wide number of analytes in the analytical scheme, to employ commercially available chemicals and reagents, to choose analytical techniques that result in quantitative and confirmatory results, and to choose the minimum possible sample consumption according to the sensitivity requirements. Based on their practicality, it is expected that an increase of LIS systems will be developed in the future and this technique will attract the interest of even more scientists. Furthermore, it is recommended to develop metric tools that are specialized for automatic techniques for a more in-depth discrimination of their performance characteristics.

CRediT authorship contribution statement

Natalia Manousi: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Justyna Płotka-Wasyłka:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Erwin Rosenberg:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation. **Aristidis Anthemidis:** Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

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